Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/28/2023
PI Name:	Yamamoto, Namiko Ph.D.		
Project Title:	Micromechanics-Based Modeling of Cement Solidified in the Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	nuy12@psu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-417-1989
Organization Name:	Pennsylvania State University		
PI Address 1:	Aerospace Engineering		
PI Address 2:	556 White Course Drive		
PI Web Page:			
City:	University Park	**State:**	PA
Zip Code:	16802-1401	**Congressional District:**	12
Comments:			
Project Type:	Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2020 Physical Sciences NNH20ZDA014N: Use of the NASA Physical Sciences Informatics System – Appendix G
Start Date:	11/01/2021	**End Date:**	10/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Jackson, Enrique	**Contact Phone:**	
Contact Email:	enrique.m.jackson@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Postdoctoral associate Vishnu Saseendran, Ph.D., joined this project in May 2022, and his got promoted to Assistant Research Professor as of May 2023.		
COI Name (Institution):	Vishnu, Saseendran (Pennsylvania State University)		
Grant/Contract No.:	80NSSC22K0083		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The research goal of this proposal is to advance knowledge about the mechanical properties of cement binder fabricated in the microgravity environment using micromechanics-based modeling, based on the scanning electron microscopy (SEM) images and other materials data on the NASA Physical Science Informatics (PSI) database. As a crucial component to build shelters and habitats on the surface of the Moon and Mars using on-site resources, common cement binder formation in the microgravity environment was studied in the investigation titled as “Microgravity Investigation of Cement Solidification (MICS)” by Professor Aleksandra Radlinska’s group at Penn State [Ed. Note: see References]. Inspection of such cement, in comparison with the reference cement samples formed on the ground, indicated that the microgravity largely affected the microstructure formation: higher porosity, and larger or well-grown portlandite phase. The effect of this microstructure change on mechanical properties needs to be studied as these cements are aimed for structural applications, but such characterization is a challenge with these particular samples due to their sample porosity and small size. In this work, the Principal Investigator proposes to use the micromechanics analysis code with generalized method of cells (MAC/GMC) incorporated with the crack band model, to predict the structure-property relationship of these unique cement samples. The representative volume elements for the modeling will be created based on the SEM images from the NASA PSI database, and materials phases and associated mechanical properties will be assigned based on their X-ray powder diffraction (XRD) data also from the NASA PSI database. The broader impacts of this research proposal are 1) establishment of simulation processes to model mechanical and fracture behaviors of the cement samples from SEM images and 2) understanding about the effect of microgravity on microstructures and their mechanical properties of the cement. This study will provide the first reference simulation data for the cement binders fabricated in the microgravity environment. Such data can be compared with the currently limited experimental data to verify the structure-property trend and mechanisms about the cement samples prepared in microgravity. This study will also provide some guidance to update original ingredient mixtures and the hydration process (water-to-cement ratio, time, etc.) to improve mechanical properties of the cements formed in microgravity. Cement and concrete formation in space will be critical to form safe and economical habitats, and is expected to be in demand with the course of NASA’s Artemis program. In addition, the same or similar simulation process can be applied to study other cements and/or concrete materials formed, especially when those samples cannot be mechanically tested. References Moraes Neves, J., P.J. Collins, R.P. Wilkerson, R.N. Grugel, and A. Radlinska, "Microgravity effect on microstructural development of tri-calcium silicate (C3S) paste." Frontiers in Materials, Vol.6, No., 2019. pp. 83. Neves, J.M., "Advancing the Processing of Cement-based Materials in Extraterrestrial Environments," in Civil Engineering. 2020, The Pennsylvania State University: University Park, PA. Collins, P.J., "Microgravity's Influence on the Hydration of Tricalcium Aluminate and Gypsum Pastes," in Civil Engineering. 2019, The Pennsylvania State University: University Park, PA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The broader impacts of this study are 1) establishment of simulation processes to model mechanical and fracture behaviors of the cement samples from SEM images, and 2) understanding about the effect of microgravity on microstructures and their mechanical properties of the cement. This study will provide some guidance to update original ingredient mixtures and the hydration process (water-to-cement ratio, hydration time, etc.) to improve mechanical properties of the cements formed in microgravity. Cement and concrete formation in space will be critical to form safe and economical habitats, and is expected to be in demand with the course of NASA’s Artemis program. In addition, the same or similar simulation process can be applied to study other cements and/or concrete materials formed, especially when those samples cannot be mechanically tested.
	In this work, the Principal Investigator (PI) proposed to use the micromechanics analysis code with generalized method of cells (MAC/GMC), to predict the structure-property relationship of these unique cement samples. First, the sparse data of cement hydrated in microgravity environment is converted to high-fidelity 3D microstructure with the aid of a deep learning-based microstructure methodology; this work was reported last year. The reconstructed sub-volumes are compared against micro-CT data using statistical descriptors. Second, the sub-volumes (partitioned based on individual constituents) of such reconstructed 3D structures are fed into the NASA Multiscale Analysis Tool (NASMAT) to obtain homogenized properties of the cement. A semi-automated Python script was developed to complete the workflow. Owing to the high computational demand, the analyses are carried out in NASA Advanced Supercomputing (NAS) servers. If successful, this project will provide the first reference simulation data for cement fabricated in the microgravity environment. The key activities carried out in this past fiscal year, under the support from this grant, include: 1. Inspection of ground-based (1g) and space-returned (μg) cement samples using X-ray microcomputed tomography (micro-CT). 2. Analysis of micro-CT virtual samples and subsequent comparison of reconstructed sub-volumes using statistical descriptors. 3. Fabrication of 1g samples to produce limited reference experimental characterization. 4. Presentation and publication: a. Part of the work titled, “Reconstruction of Tricalcium Silicate Microstructures for Repeating Unit Cell Analysis” was presented at the AIAA SciTech forum 2023. b. Manuscript titled, “Unlocking the Potential: Analyzing 3D Microstructure of Small-scale Cement Samples from Space using Deep Learning” was submitted to the npjMicrogravity journal. [Ed. Note: See Bibliography for complete citations.] 1. X-ray Micro-CT Imaging The samples fabricated on ground (referred to as 1g sample from now on) and the samples cured aboard the International Space Station (ISS) (referred to as μg sample from now on) were inspected by microCT scans, using a Zeiss Xradia 620 Versa X-ray Microscope at the Pennsylvania State University (Penn State) Energy and Environmental Sustainability Lab. For both samples, a specimen size ~ 10 × 6 × 6 mm was extracted and fixed atop a specimen mount. Scans were performed at 50 kV and 4s exposure time. The total scan time was 2.5 h and the data were acquired with a 4x optical magnification. In total, each virtual sample contained ~ 999 million voxels with a spatial resolution of ~ 2 μm. The 3D virtual samples were generated using 16-bit cross-section images from the microCT scan (996

Task Progress:	through-thickness slices). For ease of data handling and post processing, the images were reduced to 8-bit [0 - 255 range]; wherein, 0 (black) represents porosity, and 255 (white) represents a solid phase. Micro-CT 3D volumes were obtained by stacking 2D cross-sectional images along the through-thickness direction. Additionally, contrast enhancement, along with a noise-smoothing filter, was applied. The threshold bounds to discern each phase was determined from the porosity, CH and C-S-H estimates, using a coupled overflow method and local-minima approach. This post processing was performed by selecting a region of interest (ROI) with the desired edge length. The micro-CT reconstruction and volumetric analysis were carried out in Dragonfly software. 2. Validation of Reconstructed Microstructure Sub-volumes by MicroCT Scans The 3D 1g and µg microstructures reconstructed by CNN were assessed by comparing them with the 3D virtual data obtained from microCT scans. The 3D microstructure data, obtained by CNN-reconstruction from 2D SEM images and by microCT scans were compared. The sub-volume was extracted from micro-CT data to compare against the 3D volumes reconstructed using 512 x 512 pixels exemplar (resolution 0.54 µm). In addition to the whole microstructures, the portlandite phase and porosity were shown, as these two phases capture the differences between the 1g and µg samples. To note, the portlandite phase is uniformly distributed with the 1g sample, while the portlandite phase shows the distinct plate-like morphology within the µg sample, in both CNN-reconstructed and microCT-inspected data. The reconstructed 3D volumes of both samples show similar microstructural characteristics to the 2D target image. The 3D microstructure data, obtained by CNN-reconstruction from 2D SEM images and by microCT scans were compared qualitatively, using the statistical descriptors. Quantitative comparison was made by using the low-order probability distribution functions: two-point correlation function, S2(r), two-point cluster function, C2(r), and lineal-path function; L2(r) can be used to fully describe microstructure characteristics. For both CNN-reconstructed and microCT-inspected data, sub-volumes were extracted and used for this analysis. In the case of the ground-based 1g sample, the synthesized volumes using exemplars matched well with the micro-CT virtual volumes. Invariably, in texture synthesis models, the efficacy of the reconstruction is dependent on the inputted target image. For the given resolution of 0.54 µm/pixel, at least an exemplar size of 512 x 512 pixels is recommended for both 1g and µg samples. 3. Modeling of Mechanical Properties by NASMAT– 1g Sample After confirming the validity of the reconstructed sub-volumes, a workflow was created to seamlessly integrate such sub-volumes into NASMAT as Repeating Unit Cells (RUCs), to be solved using the High-Fidelity Generalized Methods of Cells (HFGMC) theory. The challenge we have faced has been the memory limitations to model the reconstructed RUC with the edge length of 276.4 µm (512 × 512 × 512 voxels, resolution 0.54 µm/pixel) using the NASMAT. Thus, the given volume was broken down into N sub-volumes, each comprised of m × m × m voxels. This strategy aided in overcoming the memory limitations and the elasticity matrix was calculated for a given volume of the 1g sample. The 1g samples were analyzed by discretizing Level-0 as 32 and Level-1 as 16 sub-cells. The input Young's modulus of the phases - portlandite, CH and C-S-H matrix were taken as 35 GPa, and 21.7 GPa, respectively. A very low Young's modulus value (~ 1 MPa) was considered for the porosity. The stiffness matrix was obtained from NASMAT.A high level of orthotropy was observed with E22 < E11, E33. Furthermore, Rule of Mixtures (ROM) analysis was conducted to verify the numerical prediction, which yielded E = 14.60 GPa (isotropic) for the hydrated C3S sample. A difference of 8.9 % was noted between E11 and ROM. The analysis was performed on the NAS cluster and took 23h to complete. A similar approach will be adopted to extract elastic constants of the µg sample using a RUC of size 512 × 512 × 512 voxels. In its current version, NASMAT is incapable of analyzing large volumes; for instance, the largest reconstructed RUC using a 1g / µg Backscattered Electron (BSE) exemplar is 512 × 512 × 512 voxels. Therefore, currently the focus is to reduce the computational resources by performing a hierarchical-based analysis, by exploiting the recursive nature of the NASMAT code. A novel hierarchical analysis that also captures the length scales is currently in progress. A schema was developed to perform a convergence study, wherein the given microstructural volume is sub-divided into various tiers. Such a study, utilizing reconstructed 1g and µg RUCs, paves the way in addressing the required partitioning of scales in hierarchical micromechanics-based analysis for highly porous 3D multiphase materials.
Bibliography Type:	Description: (Last Updated: 09/26/2024)
Articles in Other Journals or Periodicals	Saseendran V, Yamamoto N, Collins P, Radlinska A, Mueller S, Jackson EM. "Unlocking the potential: Analyzing 3D microstructure of small-scale cement samples from space using deep learning." npj Microgavity, under review as of September 2023. , Sep-2023
Papers from Meeting Proceedings	Saseendran V, Yamamoto N, Collins P, Radlinska A, Pineda EJ, Bednarcyck BA. "Reconstruction of Tricalcium Silicate Microstructures for repeating unit cell analysis." AIAA SciTech 2023 Forum, p. 2025. 2023, National Harbor, MD, January 23-27, 2023. Abstracts. AIAA SciTech 2023 Forum, p. 2025. 2023, National Harbor, MD, January 23-27, 2023. , Jan-2023